In [1]:
```
import pandas as pd
import numpy as np
import matplotlib
```

In [22]:
```
df = pd.read_csv('Phase_1.csv')
```

In [23]:
```
df
```

Out[23]:

	Invoice (F)	StockCode	Description	Quantity	InvoiceDate (R)	Price	Customer ID	Coun
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	12-01-2009 07:45	6.95	13085	Uni Kingd
1	489434	79323P	PINK CHERRY LIGHTS	12	12-01-2009 07:45	6.75	13085	Uni Kingd
2	489434	79323W	WHITE CHERRY LIGHTS	12	12-01-2009 07:45	6.75	13085	Uni Kingd
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	12-01-2009 07:45	2.10	13085	Uni Kingd
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	12-01-2009 07:45	1.25	13085	Uni Kingd
...	...	...	...	...	...	...	...	
8335	490310	21551	CERAMIC BIRDHOUSE BUTTERFLY LARGE	6	12-04-2009 14:56	5.95	17151	Uni Kingd
8336	490310	21550	CERAMIC BIRDHOUSE CRESTED TIT LARGE	12	12-04-2009 14:56	5.45	17151	Uni Kingd
8337	490310	21552	CERAMIC BIRDHOUSE BLACK ROSE LARGE	12	12-04-2009 14:56	5.45	17151	Uni Kingd
8338	490310	21554	CERAMIC BIRDHOUSE RED ROOF LARGE	12	12-04-2009 14:56	5.45	17151	Uni Kingd
8339	490311	21232	STRAWBERRY CERAMIC TRINKET BOX	12	12-04-2009 15:11	1.25	16451	Uni Kingd

8340 rows × 13 columns

In [ ]:

```
#REMOVING UNAMED COLUMNS
```

In [27]:

```
df = df.loc[:, ~df.columns.str.contains('^Unnamed')]
df #Removed all the unamed column
```

Out[27]:

	Invoice (F)	StockCode	Description	Quantity	InvoiceDate (R)	Price	Customer ID	Coun
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	12-01-2009 07:45	6.95	13085	Uni Kingd
1	489434	79323P	PINK CHERRY LIGHTS	12	12-01-2009 07:45	6.75	13085	Uni Kingd
2	489434	79323W	WHITE CHERRY LIGHTS	12	12-01-2009 07:45	6.75	13085	Uni Kingd
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	12-01-2009 07:45	2.10	13085	Uni Kingd
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	12-01-2009 07:45	1.25	13085	Uni Kingd
...	...	...	...	...	...	...	...	
8335	490310	21551	CERAMIC BIRDHOUSE BUTTERFLY LARGE	6	12-04-2009 14:56	5.95	17151	Uni Kingd
8336	490310	21550	CERAMIC BIRDHOUSE CRESTED TIT LARGE	12	12-04-2009 14:56	5.45	17151	Uni Kingd
8337	490310	21552	CERAMIC BIRDHOUSE BLACK ROSE LARGE	12	12-04-2009 14:56	5.45	17151	Uni Kingd
8338	490310	21554	CERAMIC BIRDHOUSE RED ROOF LARGE	12	12-04-2009 14:56	5.45	17151	Uni Kingd
8339	490311	21232	STRAWBERRY CERAMIC TRINKET BOX	12	12-04-2009 15:11	1.25	16451	Uni Kingd

8340 rows × 9 columns

In [6]:

```
#CHECKING NULL VALUES
```

In [28]:

```
df.isnull().any() #No Null Values Found
```

Out[28]:
```
Invoice (F)         False
StockCode           False
Description         False
Quantity            False
InvoiceDate (R)     False
Price               False
Customer ID         False
Country             False
TotalAmount (M)     False
dtype: bool
```

In [ ]:
```
#CHECKING NEGATIVE QUANTITY
```

In [29]:
```
(df['Quantity']<0).sum()
```

Out[29]:
```
np.int64(0)
```

In [42]:
```
df[df['Quantity']<0]
```

Out[42]:

Invoice (F)	StockCode	Description	Quantity	InvoiceDate (R)	Price	Customer ID	Country	To

In [ ]:
```
#CHECKING ZERO PRICE
```

In [24]:
```
#df.to_excel("cleaned_data_phase_1.xlsx", index=False) #Imported the file to xls
```

In [31]:
```
(df['Price']==0).sum()
```

Out[31]:
```
np.int64(0)
```

In [39]:
```
df[df['Price']==0]
```

Out[39]:

Invoice (F)	StockCode	Description	Quantity	InvoiceDate (R)	Price	Customer ID	Country	To

In [33]:
```
(df['Price']==0).any() #False #Zero Price Checked
```

Out[33]:
```
np.False_
```

In [ ]:
```
#CHECKING SMALL VALUES LIKE 0.000001
```

In [40]:
```
df[df['Price']<=0] #No Small Values Found
```

Out[40]:

Invoice (F)	StockCode	Description	Quantity	InvoiceDate (R)	Price	Customer ID	Country	To

In [ ]:
```
#CHECKING NEGATIVE PRICE
```

In [35]:
```
(df['Price']<0).sum()
```

Out[35]: np.int64(0)

In [38]:
```
df[df['Price']<0] #No Negative Price Found
```

Out[38]:

Invoice (F)	StockCode	Description	Quantity	InvoiceDate (R)	Price	Customer ID	Country	To